VERSION
14.0



Assembly

AFLCMC/WA

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AMS GRA ARCHITECTURE VOLUMES

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1.0 Overview

The Assembly provides an entry point for adopters who do not have and are not procuring an Agile Mission Suite (AMS) Government Reference Architecture (GRA)-compliant Mission System. It provides a mechanism to procure new or upgraded capabilities, aligned to the functional partitioning boundaries defined by AMS GRA. Sample procurement use cases that might use an Assembly include transitioning a legacy deployed system to an AMS GRA-compliant Mission System, certifying low-Size, Weight, and Power (SWAP) systems which do not fit the definition of a full Mission System, or fielding a system with hardware not well aligned to AMS GRA Minimum Procurable Unit (MPU) boundaries.

An Assembly must include at least one AMS GRA-compliant Service MPU and may additionally include legacy Mission System components, available AMS GRA-compliant hardware MPUs, or other AMS GRA-defined elements (e.g., Mission Agnostic Service Infrastructure (MASI) services).

An Assembly will instantiate a software environment and AMS GRA interfaces that facilitate AMS Service portability. If an Assembly element communicates with an AMS GRA-compliant MPU, that MPU must be part of the Assembly.

1.1 Purpose

The Assembly Volume describes the physical and logical aspects of an AMS GRA-compliant Assembly. Furthermore, this Volume provides details on architectural considerations likely needed by an Assembly but not wholly specified by AMS GRA.

The purpose of the Assembly Volume is to provide guidance to a Mission System or Assembly Integrator on how to best procure and integrate new or upgraded capabilities into AMS GRA-compliant Assemblies and Mission Systems. Those entities interested in a partially AMS GRA-compliant Mission Systems should first consult the AMS GRA architectural volume for the Mission System.

The volume also provides compliance requirements against which an AMS GRA-compliant Assembly is assessed.

1.2 Scope

The Assembly Volume is applicable to entities seeking AMS GRA compliance or performing a compliance assessment of a Assembly.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority or system integrator refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA Assembly is compliant when it provides the required prescriptive constructs detailed in §3. See **Table 1.3.1** for a list of authoritative references for the Assembly Volume.

Standard	Title	Version/Date
AMS GRA Volumes	MASI Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Time Distribution Volume	14.0
AMS GRA Volumes	AMS GRA Service MPU Volume	14.0
AMS GRA Volumes	CDS MPU Volume	14.0
AMS GRA Volumes	Countermeasures Manager (CMM) MPU Volume	14.0
AMS GRA Volumes	Cryptographic Unit MPU Volume	14.0
AMS GRA Volumes	DPP MPU Volume	14.0
AMS GRA Volumes	Data Storage MPU Volume	14.0
AMS GRA Volumes	Local Function MPU Volume	14.0
AMS GRA Volumes	MFA MPU Volume	14.0
AMS GRA Volumes	OMS Service MPU Volume	14.0
AMS GRA Volumes	PCaB MPU Volume	14.0
AMS GRA Volumes	Switch MPU Volume	14.0
AMS GRA Models	AMS GRA Architecture Model	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the Assembly.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the Assembly Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Software Architecture Volume	14.0
AMS GRA Volumes	Security Framework Volume	14.0
AMS GRA Volumes	Resource Management Volume	14.0
AMS GRA Volumes	Pilot Vehicle Interface Volume	14.0
AMS GRA Volumes	Executive Summary	14.0
AMS GRA Volumes	Cyber IDD	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the Assembly and Related Entities.

2.0 Assembly Definition

The AMS GRA compliance tiers for Mission Systems provide an approach that supports the architectural and business goals of AMS GRA while designing, developing, and integrating MPUs into a platform. The Assembly construct is intended to support alternative use cases for Mission System procurement, situating the AMS GRA Assembly as a site to merge existing and new AMS GRA-compliant capabilities into a deployable system that permits non-compliant components.

An Assembly composed of some legacy Mission System elements is a useful platform for integrating AMS GRA software MPUs, but may also serve use cases where legacy elements are replaced by compliant hardware MPUs.

2.1 Assembly

An Assembly must include at least one AMS GRA-compliant Service MPU. Required characteristics for an AMS GRA-compliant Assembly are described in **Table 3.0.1**.

An Assembly may incorporate legacy non-compliant Mission System hardware elements. An adopter implementing a full complement of AMS GRA MPUs should consider implementing a Mission System rather than an Assembly.

Given that any AMS GRA-compliant MPUs contained in the assembly are required to communicate over an implementation of AMS GRA interfaces, the Assembly shall include necessary implementation of AMS GRA interfaces that facilitate communications between AMS GRA-compliant Assembly MPUs.

An Assembly will instantiate a software environment and AMS GRA interfaces that facilitate AMS Service portability. If an Assembly element communicates with an AMS GRA-compliant MPU, that MPU must be part of the Assembly.

3.0 Compliance Requirements

Table 3.0.1 describes the required Assembly characteristics. Where "Mission System" appears in the text of these requirements, it should be interpreted as "Assembly" in the context of an AMS GRA Assembly Compliance Assessment. These requirements are a subset of the full list allocated to an AMS GRA Mission System and MASI. See the Mission Agnostic Service Infrastructure (MASI) Volume guidance for designing and implementing the allocated MASI requirements.

Although the identified requirements have associated compliance tiers in the MASI and Mission System contexts, specific tiering for the Assembly is still under consideration and may change in the future.

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MASI-001	A MASI shall be able to start mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration
GRA-MASI-002	A MASI shall be able to stop mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration
GRA-MASI-003	A MASI shall monitor the Management Interface for reported health and status of AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-006	A MASI shall monitor fault information reported by AMS GRA-compliant MPUs.	Tier 1	Inspection
GRA-MASI-008	A MASI shall consume all available BIT information reported by AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-015	A MASI shall monitor the health and status reporting of all deployed AMS GRA and OMS Services.	Tier 1	Inspection
GRA-MASI-022	A MASI shall deploy OCI-compliant containerized software utilizing a container framework.	Tier 1	Demonstration
GRA-MASI-024	A MASI shall facilitate endpoint connections between services.	Tier 1	Demonstration
GRA-MS-005	A Mission System shall implement access and information flow isolation between safety critical and non-safety critical processing.	Tier 1	Analysis
GRA-MS-007	A Mission System shall implement an API on DPPs, as applicable, to allow AMS GRA Services and OMS Services to access OMS Special Signals.	Tier 2	Inspection
GRA-MS-008	A Mission System shall implement an API on DPPs, as applicable, to allow AMS GRA Services and OMS Services to access OMS platform-specific Elements.	Tier 2	Inspection
GRA-MS-010	A Mission System shall provide an AMS GRA-compliant MASI implementation.	Tier 1	Inspection
GRA-MS-013	A Mission System that segregates data into multiple security domains shall utilize an approved Transfer CDS to pass data between security domains.	Tier 3	Analysis
GRA-MS-014	A Mission System utilizing an MLS bus shall enforce logical segregation of security domains by utilizing an approved MLS CDS at each bus interface.	Tier 3	Analysis
GRA-MS-023	A Mission System shall make available all MEL Implementations supplied by MFA providers to any AMS GRA Service to be loaded.	Tier 1	Analysis
GRA-MS-026	A Mission System shall make available to MASI the locality between DPPs, and DPPs to MFAs to inform the deployment and recovery strategy.	Tier 2	Inspection
GRA-MS-027	A Mission System shall provide the capability of all AMS GRA Services to be deployed to any ABB-connected compatible DPP.	Tier 1	Analysis
GRA-MS-028	A Mission System employing MLS logical separation on an MLS bus shall provide interoperable authentication and integrity mechanisms consistent for communicating endpoints in an approved manner.	Tier 3	Analysis

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-029	The Mission System shall provide open documentation for the incorporated MLS solution regarding programming of security policies and switch configuration if multiple security domains or security policies are present in support of approved certification.	Tier 3	Inspection
GRA-MS-030	A Mission System shall include non-proprietary documentation for interfaces operating at multiple security domains.	Tier 3	Inspection
GRA-MS-031	A Mission System shall include MLS interface documentation with descriptions of boundaries and allowable data flows between multiple interconnected security domains.	Tier 3	Inspection
GRA-MS-032	A Mission System shall include interface documentation for hardware requirements and interface design details when multiple interconnected security domains are present within the Mission System.	Tier 3	Inspection
GRA-MS-033	A Mission System with classified data shall encrypt all classified DaR using approved cryptographic algorithms.	Tier 1	Inspection
GRA-MS-034	A Mission System shall include a model consistent with MBSE content contained in the AMS GRA SK release with which the Mission System is compliant.	Tier 2	Inspection
GRA-MS-035	A Mission System shall use approved cryptographic algorithms to authenticate and verify the integrity of all Mission System data, software and firmware prior to updating, loading, and executing.	Tier 3	Inspection
GRA-MS-036	The Mission System documentation shall document the implementation of the approved cryptographic authentication and verification algorithms for all loadable Mission System data, software and firmware prior to updating, loading, and executing.	Tier 3	Inspection
GRA-MS-037	A Mission System shall generate audit logs for security relevant events to include generation of audit logs, response to audit failure, protection of audit information and enforcement of non repudiation.	Tier 2	Inspection
GRA-MS-041	A Mission System shall provide storage for Local Functions to be accessible to the associated MFA using networked Data Storage, non-volatile MFA local storage, or non-volatile DPP local storage.	Tier 2	Inspection
GRA-MS-045	A Mission System shall include SAST results for all microprocessor software source code in the Mission System, IAW the Mission System documentation template.	Tier 1	Inspection
GRA-MS-046	A Mission System shall include DAST results for all microprocessor software in the Mission System, IAW the Mission System documentation template.	Tier 1	Inspection
GRA-MS-049	A Mission System with CPI in non-volatile memory shall prevent recovery of CPI and protective keys in the event of a loss of Mission Platform.	Tier 1	Analysis

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-050	A Mission System shall provide interface connectivity for Required MPU interfaces.	Tier 1	Inspection
GRA-MS-051	A Mission System shall provide configurability for audit of security relevant events.	Tier 2	Inspection
GRA-MS-052	A Mission System shall securely deploy software containing CPI.	Tier 1	Demonstration
GRA-MS-054	A Mission System shall include documentation on the deployment and recovery strategy for its software elements.	Tier 1	Inspection
GRA-MS-055	A Mission System shall include non-proprietary machine readable documentation for the security audit log format.	Tier 2	Inspection
GRA-MS-056	A Mission System with MFAs shall provide an implementation of the Resource Management Framework for management of sensor resources.	Tier 1	Inspection
GRA-MS-057	The Mission System shall provide a RoT to Mission System MPUs.	Tier 2	Analysis
GRA-MS-061	The Mission System shall authorize access to endpoints on a per-session basis.	Tier 2	Analysis
GRA-MS-062	The Mission System shall establish secure communications between endpoints using NSA-approved cryptographic methods.	Tier 2	Inspection
GRA-MS-065	The Mission System shall only allow authorized communication channels IAW a file identified and documented by the Mission System.	Tier 2	Demonstration
GRA-MS-068	The Mission System shall protect security functions from unauthorized access or modification.	Tier 1	Demonstration
GRA-MS-069	The Mission System shall protect security relevant data from unauthorized access or modification.	Tier 1	Demonstration
GRA-MS-073	The Mission System shall use the MEL Implementation supplied by the MFA provider to establish the interface between an AMS GRA Service and the MFA.	Tier 1	Demonstration

Table 3.0.1: An Assembly's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
BIT	Built-In Test
CDS	Cross Domain Solution
CMM	Countermeasures Manager
CPI	Critical Program Information
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DPP	Digital Payload Processor
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
MASI	Mission Agnostic Service Infrastructure
MBSE	Model-Based Systems Engineering
MEL	Multi-Function Aperture (MFA) Encapsulation Layer
MFA	Multi-Function Aperture
MLS	Multi-Level Security
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NSA	National Security Agency
OCI	Open Container Initiative
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
PVI	Pilot Vehicle Interface
RoT	Root of Trust
SAST	Static Application Security Testing

SK	Starter Kit
SWAP	Size, Weight, and Power

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
–	Initial release	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Assembly Volume Between Documentation Releases.